

Rama Chandra Bhogasamudram

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EDUCATION

University of Southern California, Los Angeles, CA **Jan 2025 – Dec 2026(Anticipated)**
MS in Computer Engineering **GPA: 3.86**

Coursework: Reinforcement Learning and Large Language Models, Deep Learning, Hardware Foundations of Machine Learning, Machine Learning, Computer Systems Organization, Graduate Math - Probability & Statistics, Graduate Math - Linear Algebra.

Jawaharlal Nehru Technological University, India

Bachelor in Electronics and Communication Engineering

Coursework: Data Structures & Algorithms, IoT, AI/ML, Computer Architecture, VLSI

SKILLS

High-Level languages: C, Java, Python, SQL, Verilog HDL, Bash

Frameworks & Libraries: Spring, Flask, pandas, numpy, scikit-learn, PyTorch, OpenCV, Apache Kafka, Langchain

Tools & Technologies: Linux(RHEL), Virtualization, Docker, Xilinx Vivado, Anaconda, CI/CD, Splunk, REST API, Git.

EXPERIENCE

Course Grader | EE 503 - Probability for Engineers | USC **May 2025 - Aug 2025**

Apple Inc. Contractor | Tata Consultancy Services (TCS), India

Data Scientist | Python, numpy, pandas, scikit-learn, Tesseract, Langchain, MilvusDB **Mar 2023 - Jun 2024**

- Developed and deployed RAG-based APIs to improve the accuracy and efficiency of extracting key information from Apple invoice documents for enterprise & strategic analytics and budget monitoring. Derived content embeddings from the documents and implemented LLM-driven querying on the invoices with over 80% accuracy.

Software Development Engineer | Java, Spring, Kafka, Docker, Linux, Bash, Selenium **Jul 2020 - Feb 2023**

- Contributed to a Selenium-based backend web automation ecosystem infrastructure which seamlessly aids in automating functional and test executions on web applications, enhancing efficiency. Integrated features such as dashboard, events, and analytics.
- Spearheaded development and maintenance of Apple factory applications based on Java, Spring, Kafka, etc., running on robust RHEL servers. Resolved several show-stopper bugs to ensure smoother factory analytics operations, and developed various multithreaded workflows to fully utilize underlying VM resources, thereby boosting performance and efficiency. Collaborated through systems, network, security, and other SRE teams to ensure reliability and availability of the applications.
- **Lead offshore team** and directly reported to the Apple Manager to ensure seamless collaboration and successful deliverables.

ACADEMIC PROJECTS

Final round in two IoT Competitions | Technex, IIT Varanasi, India & i3Indya, IIT Bombay, India

- **Industrial IoT** - Built and showcased a miniature Industrial IoT prototype featuring conveyor-belts, access systems, electronic utilities - lights, Air Conditioning, etc and other manufacturing processes which are electronically controlled from the internet using nodeMcu and raspberry-pi.
- **RFID Smart Card** - Built an RFID smart card system to enable card readers to access multiple user documents per request and validation. This enables individuals to conveniently carry one card instead of several.

Daily on-premises college bus tracker using YOLO and OCR

- Integrated YOLO object detection and number plate recognition to track buses entering and leaving the university, thus allowing optimization of pick-up times and congestion.

Research published on “FPGA based navigation approach for the Visually Impaired”

- Developed a navigation assistance system for the visually impaired individuals using FPGA, ultrasonic sensors for proximity detection, and compass and GPS module for location, on which a journal is published on IJRASET.

ACHIEVEMENTS

- International Collegiate Programming Contest (ICPC) 2019, regional level.
- Tata Consultancy Services (TCS) CodeVita Season 8 (competitive coding contest), all-India rank 1082 out of 55,654.